A Dual Correction Guarantee Mechanism for Numerical Label Noise

Yaqing Guo, Lingxuan Cui, Gaoxia Jiang, Fan Lei, Senyu Hou, Hang Xu, and Wenjian Wang

***Abstract*—Label noise usually leads to the degradation of the model generalization in supervised learning, especially for deep neural networks. Although substantial studies have explored learning from noisy labels, most of them focus on classification, with limited effectiveness on noisy regression tasks. Moreover, advanced techniques for addressing numerical label noise typically discard almost all noisy examples directly or downweight them, thus resulting in a partial exploration of training data. In this work, we propose a numerical label noise correction method under a dual correction guarantee mechanism (named DCGM), aiming to address the problem of sample information loss and effectively reduce the noise level. First, we theoretically prove that DCGM can reduce the generalization error bound for noisy regression. And then, we resist label noise influence under a dual correction guarantee mechanism. (1) We determine genuine neighborhoods through feature embedding representations, rather than directly selecting nearest neighbors in the original feature space, to identify noisy labels more precisely and provide usable and accurate sample information for label correction. Then, we define the predictive uncertainty, and only those noisy labels with low uncertainty values are selectively corrected and retained. Other noisy samples are discarded. (2) We construct a robust heteroscedastic loss for performing post-correction. Samples with false corrections and unidentified noisy samples will be assigned smaller weights during the updating of network parameters, to implicitly conduct the second round of label correction. Experimental results demonstrate that DCGM can achieve state-of-the-art performance on noisy regression tasks.**

***Index Terms*—Learning with noisy labels, regression, label correction, dual correction guarantee, precise noise identification.**

I. Introduction

Owing to the recent development of the large-scale data collection technology, deep neural networks (DNNs) with highly parameterized structures have achieved remarkable success in various machine learning tasks, particularly in supervised learning. However, their success highly depends on sufficient high-quality labeled data [1], [2]. Although some inexpensive methods can be adopted as alternates to annotate labels, they always result in noisy labels, due to insufficient information or inexperienced workers. Obviously, because of the strong capacity for fitting any function, DNNs easily overfit these noisy data and thus lead to poor generalization performance [3], [4]. Hence, for supervised learning, learning with label noise is critical and challenging in data-centric deep learning.

Existing solutions mainly use noise-robust modeling and noisy data cleaning to deal with label noise [4], [5], [6], [7]. Unlike noisy classification tasks that have been widely discussed, regression problems corrupted by label noise are underexplored. The fundamental reason is that numerical label noises in regression tasks are more intractable. Specifically, the predicted values for continuous labels can take any numerical value within a wide range. In contrast, predicted categorical labels are restricted to specific sample classes [8]. However, regression problems cover many practical applications in real-world scenarios, such as age estimation, gaze estimation from face images, health metrics prediction in medical fields for disease diagnosis and progression tracking, etc. In addition, the popular cluster assumption in classification that similar instances should have the same label does not naturally hold for regression tasks. Hence, directly applying existing techniques for noisy classification tasks to address numerical label noise may lead to a mismatch with the data distribution, and the impact of noise is still not effectively resisted [1], [9], [10], [11].

In recent works on noisy regression tasks, noise-robust models are constructed primarily via sample weighting. After filtering identified noises, noisy data cleaning learns models on remaining clean data. It is more generic and flexible, and has shown certain performance advantages in learning from noisy labels [1], [12], [13]. Note that early filters lack the adaptability to determine filtration intensity based on actual noise levels. Instead, they only pursue lower noise levels, which may lead to overcleaning. Jiang et al. theoretically validated the critical importance of maintaining an adequate sample size for learning from noisy labels. They adopt the optimal sample selection (OSS) framework to find an appropriate balance between anti-overcleaning and effective noise removal [4], [8]. Even so, the sample size has still been significantly reduced, primarily due to inaccurate noise recognition and the subsequent direct discarding of noisy samples, thus resulting in only a partial exploration of training data. Notably, a small number of remaining training samples may cause a shift in data distribution, fail to fully reflect the underlying data distribution, and thus prevent

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Yaqing Guo, Lingxuan Cui, Gaoxia Jiang, Fan Lei, Senyu Hou, and Hang Xu are with the School of Computer and Information Technology, Shanxi University, Taiyuan 030006, China (e-mail: yqguo@sxu.edu.cn; cuilingxuan@sxu.edu.cn; jianggaoxia@sxu.edu.cn; leifan@sxu.edu.cn; housenyu@sxu.edu.cn; hxu@sxu.edu.cn).

Wenjian Wang is with the Key Laboratory of Data Intelligence and Cognitive Computing, Shanxi University, Taiyuan 030006, China (e-mail: wjwang@sxu.edu.cn).

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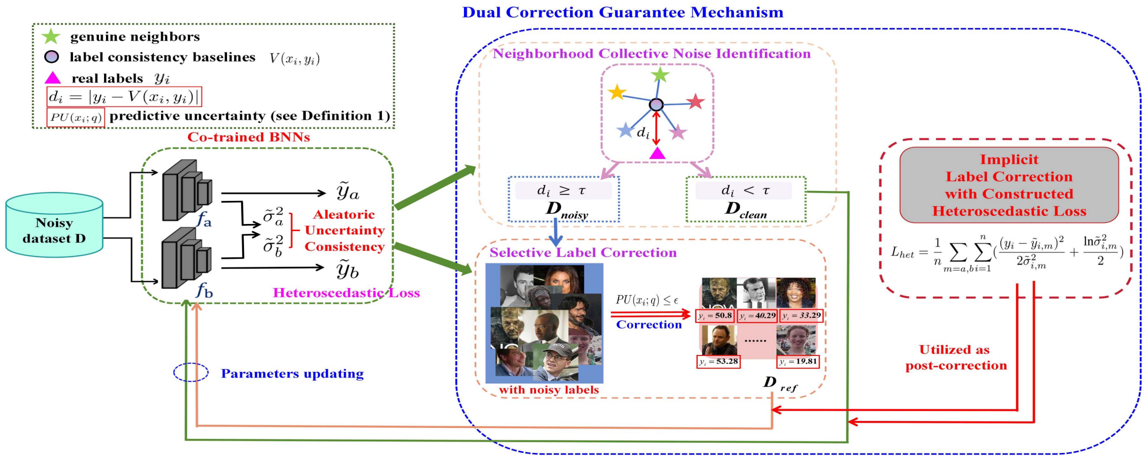


Fig. 1. Overview of the DCGM framework. Our method, DCGM, accomplishes numerical label correction through the dual correction guarantee mechanism. (1) After accurately identifying label noise based on the genuine nearest neighborhood relationships and the Gaussian mixture model, DCGM only selectively corrects noisy labels with low predictive uncertainty. (2) We construct a robust heteroscedastic loss to implicitly facilitate the second round of label correction. Under the post-correction guarantee regime, DCGM successfully boosts the quality of numerical label correction by assigning smaller weights to samples with false corrections and unidentified noisy samples. As training epochs progress, the generalization ability of Bayesian neural networks (BNNs) and the accuracy of data processing are interactively enhanced within the DCGM framework.

the effective discovery of latent patterns within the data during model learning [14], [15], [16], [17].

For more robust learning with numerical label noise, one can utilize noisy examples through label correction rather than simply discarding them, enabling the thorough exploitation of training data. Label correction is designed to maintain the original data distribution while effectively reducing the level of label noise [14], [15], [16], [17]. However, in noisy regression tasks, since the sample labels are continuous and the number of possible prediction values is unlimited, it is far more difficult to correct noisy labels accurately compared to classification tasks. Undoubtedly, improper correction for label noise may introduce additional label noise to model learning. In such cases, label correction not only fails to effectively mitigate the impact of noise, but also further deteriorates the model performance. Given the aforementioned difficulty, to the best of our knowledge, very limited efforts have been devoted to exploring label noise correction for regression problems, unlike for classification tasks. Achieving high-quality numerical label correction for regression tasks remains an extremely challenging task.

In order to effectively correct numerical label noise, four key issues require our rigorous investigation: (1) Precisely filtering out noisy samples provides accurate sample information for label correction. (2) Clearly define the selection criteria for corrigible noisy samples to prevent the introduction of incorrect label corrections. (3) Determine reasonable label correction values for numerical label noise. (4) Design a scientific correction process to iteratively improve the accuracy of each module involved in learning noisy labels.

In this paper, we propose a novel numerical label noise correction method under a dual correction guarantee mechanism (named DCGM). To improve generalization performance, after ensuring the correct filtering of noisy samples, DCGM selectively corrects label noise under a post-correction guarantee regime in each epoch, as shown in Fig. 1. DCGM can achieve high-quality numerical label correction via the dual correction guarantee mechanism, thus, it is applicable to noisy regression problems. The main contributions are as follows:

- We theoretically demonstrate that DCGM can reduce the generalization error bound for learning from numerical label noise.
- We propose neighborhood collective noise identification, in which the true label of each candidate sample is approximated using the Gaussian mixture model (GMM) and nearest neighborhood relationships to identify label noise in a relatively unbiased manner. The nearest neighbors of each sample are found based on discriminative representations in the mapped feature space, which can capture the essential characteristics of samples, thus resulting in a notable enhancement on nearest neighbor search precision.
- We define a metric for determining whether noisy samples will be selectively corrected, so as to avoid low-quality label correction. Using identified clean samples and refurbishable noisy samples, network parameters are robustly updated with sufficient and reliable data as training epochs progress, which can ensure nearly comprehensive exploration of the training data.
- We construct a post-correction guarantee mechanism to further ensure the accuracy of numerical label correction. By developing a robust heteroscedastic loss for network parameter updating based on aleatoric uncertainty estimation, small weights are precisely assigned to samples with false corrections and unidentified noisy samples, which implicitly conducts robust label correction again. The accuracy of noise verification is also further guaranteed.

II. Related Works

A. Noise-Robust Modeling.

Noise modeling directly constructs robust models for noisy regression data, without requiring any data preprocessing [1], [8], [13], [18]. This is achieved by using commonly used strategies for robust modeling, such as robust loss design, sample weighting, regularization techniques, robust architectures, ensemble methods, etc. However, its performance is still unstable for severe noise and unknown noise. Besides, the dominant technique for constructing robust models, sample weighting, can not ensure a full exploration of the regression training data, especially at the high noise rate [1], [5], [16], [19].

B. Label Noise Filtering.

Label noise filtering mitigates the impact of label noise at the data level. It improves the generalization ability of learners by discarding samples identified as noises. For regression tasks, common noise filters can be categorized into four groups: prediction ensemble filtering, noise estimation-based filtering, threshold-based filtering, and nearest neighbor-based filtering [8]. Among them, prediction ensemble filtering distinguishes noisy samples via majority voting schemes applied to predicted labels. Its base models can be either heterogeneous models or homogeneous models trained on different subsets [20], [21]. Noise estimation-based filters first perform noise estimation by determining confidence intervals or the posterior distributions of true labels. Then, the OSS framework is employed to identify an optimal trade-off between filtering efficacy and excessive data cleaning, thus ensuring robust generalization performance [4], [22]. As for threshold-based filtering, it selects clean samples according to sample loss values or noise scores calculated by the predicted accuracy of each base model [20], [23]. Note that above filters only utilize predicted label distributions of individual candidate samples to identify label noises, which may cause the confirmation bias in the process of noise filtering. Incorrect judgment regarding the contamination state of a sample, caused by inaccurate estimation for its true label value, is known as confirmation bias [14].

Differently, nearest neighbor-based filtering achieves noise filtering by leveraging neighborhood information. It considers that clean samples should have high label similarity with their nearest neighbors [8], [14]. Regretfully, neighbor-based filters for regression can not achieve the same level of noise recognition performance as those in classification tasks. Because regression problems only hold the manifold assumption (i.e., similar instances have similar labels), the true labels of candidate samples are difficult to estimate [24]. In addition, all aforementioned filters only exploit the training set partially. They directly discard identified noisy samples, which may result in the loss of useful information for robust training [1].

C. Label Noise Correction.

Label noise correction also falls within the scope of noisy data cleaning. The key distinction between it and noise filtering lies in the fact that the former endeavors to clean noisy data by replacing corrupted labels with true labels. These true labels are obtained according to model predictions or feature representations. Moreover, considering that the quality of label corrections may vary significantly, several selective label correction methods have been proposed successively. They only correct noisy labels with low predictive uncertainty, thereby preventing error accumulation [5], [14], [15], [16], [17]. Although numerous studies have explored label correction, they are all oriented towards classification tasks, with limited effectiveness on learning form numerical label noise [1], [11]. Hence, designing a proper label correction method for regression tasks deserves our attention.This also involves exploring how to enable noise identification and label correction to mutually reinforce each other, thereby achieving more robust model learning from noisy labels.

III. The Proposed Method

Considering noisy regression data sets $D = \{x_i, y_i\}_{i=1}^{n}$, where $x_i \in R^p$ denotes the feature vector, y_i may be the label noise whose corresponding ground-truth label is $y_i{}^*$, i.e., $y_i = y_i{}^* + \varepsilon_i$, learning from noisy labels aims to enhance the generalization performance of regression models on noisy data sets D [1].

A. Model Overview

In standard deep learning, the parameters θ of the neural network is directly updated according to the descent direction of the certain loss on noisy data sets D. Unfortunately, DNNs overfit corrupted data sets easily due to the strong fitting capability, which may lead to poor generalizability in testing data sets. Although noise filtering can alleviate the above issue to some extent, it only results in a partial exploration of the training data [1], [3], [16]. DCGM modifies the above learning processes to more robustly mitigate the impact of label noise. Specifically, we first propose novel neighborhood collective noise identification to recognize label noise accurately and construct the clean set $D_C \subseteq D$. Then, to describe the training data distribution more comprehensively, we selectively refurbish noisy samples from the noisy set D_N and construct the refurbishable set D_{ref}. Note that we only correct noisy samples with low predictive uncertainty to ensure the quality of numerical label noise correction. Finally, we correct the backward loss of co-trained BNNs by using D_C and D_{ref}. The above backward loss is the heteroscedastic loss with uncertainty consistency (L_{BNN}), which is utilized as post-correction. DCGM only backpropagates heteroscedastic losses for identified clean samples and refurbishable samples to update the network parameters, i.e.,

$$\theta_{t+1} = \theta_t - \alpha\nabla\left(\frac{1}{|D_C \cup D_{ref}|}\left(\sum_{x_i \in D_{ref}} L_{BNN}(x_i, y_i^{ref}; \theta_t)\right.\right.$$
$$\left.\left. + \sum_{x_i \in D_C} L_{BNN}(x_i, y_i; \theta_t)\right)\right). \quad (1)$$

Here, the BNN is used as the network architecture. The use of the BNN is due to the following considerations. First, in the dual correction guarantee mechanism, we need to define a metric for judging whether to perform selective label correction using prediction results. Since BNNs have the ability to output the uncertainty of predicted values, they are more suitable for selective label correction than ordinary neural networks.

Second, in order to conduct the post-correction, the heteroscedastic loss for network updating should be constructed. Since BNNs can predict aleatoric uncertainty σ_i^2 of outputs jointly with the target value, in DCGM, this σ_i^2 is utilized as the noise variance parameter for the heteroscedastic loss. This mechanism implicitly corrects predicted labels through sample weighting. Note that aleatoric uncertainty σ_i^2 refers to the magnitude of label noise of each candidate. The samples exhibiting higher aleatoric uncertainty are more likely to contain larger numerical label noise [11], [12], [25]. Considering the same sample should have equal predicted aleatoric uncertainty even in different BNNs, we introduce the aleatoric uncertainty consistency into the heteroscedastic loss via the co-trained learning model, to ensure the estimation accuracy for aleatoric uncertainty [25]. Besides, co-trained BNNs with robust heteroscedastic loss are also beneficial for initial noise identification and selective label correction.

B. Neighborhood Collective Noise Identification

For label noise learning, existing research executing noise verification solely on the basis of predicted label distributions of individual samples may incur confirmation bias. This is not conducive for accurately identifying label noise [14]. The noise recognizer, which determines whether a candidate sample contains label noise or not according to the consistency between its real label and predicted label distributions of its K nearest neighbors, can effectively avoid the above confirmation bias. However, in regression tasks, it is difficult to accurately estimate the true label values of candidate samples by relying on the predicted label distributions of their K nearest neighbors. Unlike in classification tasks, where the labels of candidate samples and those of their neighboring samples tend to be highly consistent, the true label values of candidate samples in regression tasks often differ from those of their neighbors. The labels of the neighboring samples themselves may also exhibit significant discrepancies. Furthermore, the number of possible values to predict in regression is unlimited. Undoubtedly, these pose considerable challenges for identifying numerical label noise according to the deviations between real labels and their corresponding estimated true labels.

Aiming at addressing the above issue, we formulate a novel regression noise identification function that determines whether a candidate is a noisy sample or not through the estimation of its label consistency baseline. Here, the Gaussian mixture model [12] is adopted to simulate the posterior distributions of true labels of candidate samples based on predicted values of their K genuine nearest neighbors [26].

Specifically, motivated by the observation that latent embedding representations learned via nonlinear feature mapping are more discriminative for characterizing samples (task-relevant), we identify genuine neighbors by computing the cosine similarity on above discriminative feature vectorial embeddings, instead of the conventional neighbor selection based on original feature vectors, thus improving the accuracy of nearest neighbor search [26]. This can prevent misjudged neighbors from affecting the estimation of the true labels for the candidate samples. The genuine neighbors can be formulated as follows:

$$\{{z_i}^{(k)}\}, k = 1, \ldots, K, \leftarrow \mathbf{KNN}(z_i; D_{train}; K), \tag{2}$$

where $z_i = u(x_i; \theta_u)$ denotes the discriminative feature representation (vectorial embedding) of the sample-label pair (x_i, y_i) generated by the encoder network u. In the noise identification step, given a candidate sample (x_i, y_i) from corrupted data sets, we first accurately find its K nearest neighbors in the mapping feature space. In the early stage of network parameter learning, we utilize the memory effect and robust BNNs to resist the influence of label noise on the discriminative feature representation learning. Note that we do not consider the candidate sample (x_i, y_i) itself when searching for its genuine nearest neighbors.

Subsequently, the neighborhood-based label consistency baseline for noise verification is defined as follows:

$$V(x_i, y_i) = \arg\max_{\tilde{y}_i} \sum_{k=1}^{K} \pi_k \phi(\tilde{y}_i; \mu_k, \Sigma_k). \tag{3}$$

The label consistency baseline essentially refers to estimating the true labels of candidate samples based on the predicted labels of their genuine nearest neighbors. Unlike simply calculating the mean of the predicted target values of nearest neighbors, it first simulates the posterior distribution of the true label of each candidate sample based on the nearest neighbor relationship, and then obtains the true label via the maximum a posteriori (MAP) estimation and Bayes ideas [9], [27]. Since the GMM is a universal approximator for almost any distribution, we adopt the GMM to simulate the posterior distribution of the true label (see (3)) [12]. Based on neighborhood priors, the mean value of the kth Gaussian component ϕ_k, i.e., μ_k, is set to the predicted target value of the $k-$th genuine nearest neighbor of (x_i, y_i): ${y_i}^{(k)}$. $\pi_k = d_k^* / \sum_{k=1}^{K} d_k^*$ denotes the mixing proportion, where $\bar{\mu}_k$ is the mean of ${y_i}^{(k)}$ and $d_k^* = 1/|{y_i}^{(k)} - \bar{\mu}_k|$. According to Ref. [28], the standard deviation of the covariance matrix of the GMM is set to $\Sigma_k = (\frac{4}{3K})^{\frac{1}{5}} \cdot \frac{\mathrm{median}\{|{y_i}^{(k)} - \bar{\mu}_k|\}}{\Phi^{-1}(3/4)}$. These settings can mitigate the adverse effects of abnormal neighbor predictions on GMM fitting, and avoid the drawbacks inherent in the EM algorithm (for calculating GMM parameters), including global iterative optimization overhead, sensitivity to initial parameter values, and inferior performance on small sample sets. Obviously, the proposed label inconsistency baseline is less affected by the abnormal model predictions at neighbors, the manifold assumption, and the unlimited number of possible prediction values for a candidate sample. It can more accurately estimate the ground-truth labels for regression tasks, as validated by Ref. [22]. The neighborhood-based label inconsistency baseline is relatively unbiased, model predictions at genuine nearest neighbors will implicitly diversify the predictive information of each candidate sample.

According to the principle of neighborhood consistency, noisy labels will exhibit significant deviations from the model predictions of their nearest neighbors [14]. Hence, we believe that candidate samples whose real labels deviate significantly from their label consistency baseline are more likely to be noisy samples. After setting a threshold τ, we re-partition data sets into two types of samples: clean samples and noisy samples. If the real label of a candidate deviates significantly from its corresponding $V(x_i, y_i)$, i.e., $d_i = |y_i - V(x_i, y_i)| \geq \tau$, then it will be assigned to the noisy set D_N, and otherwise, the clean set D_C.

C. Dual Correction Guarantee Mechanism

Based on the partitioned data set from the noise verification step, noisy samples will be corrected by a robust label corrector. Note that neural networks still own perceptual consistency before fully fitting noisy data [15]. Currently, only noisy samples with low predictive uncertainty are selectively refurbished according to model predictions in DCGM, to avoid low-quality label correction. In addition, to acquire stable label correction, the predictive uncertainty of each noisy sample is quantified as the average of its corresponding predictive uncertainties across q epochs (i.e., the epochs that a sample undergoes after being selected as label noise). The notion of final predictive uncertainty of each noisy sample is formalized as Definition 1.

Definition 1: A noisy sample x_i will be corrected only when its average predictive uncertainty from q epochs $PU(x_i; q) \leq \epsilon$:

$$PU(x_i; q) = \frac{1}{q}\sum_{\bar{q}=1}^{q}\frac{\tilde{\sigma}^2_{(i,a;\bar{q})} + \tilde{\sigma}^2_{(i,b;\bar{q})}}{2}.$$

Note that min-max normalization is applied on the $PU(x_i; q)$ values of noisy samples in actual implementation. Consequently, $0 \leq \epsilon \leq 1$.

DCGM selectively corrects noisy labels and constructs the refurbishable set D_{ref} according to the following formula:

$$y_i^{correct} = (\hat{y}_{i,a} + \hat{y}_{i,b})/2, \text{if } x_i \in \mathcal{D}_{ref}, \tag{4}$$

where $\hat{y}_{i,m}$ denotes the average of q predictions from q epochs of a single BNN, i.e., $\hat{y}_{i,m} = \frac{1}{q}\sum_{\bar{q}=1}^{q}\tilde{y}_{i,m}$. As for the remaining samples of the current noisy set, which may accumulate the model error, the proposed method excludes them from network parameters updating, to pursue robust regression learning.

Considering the difficulty of accurately correcting numerical label noise, we introduce a post-correction guarantee regime into selective label correction. In DCGM, we construct the robust heteroscedastic loss for implicitly correcting the refurbishable results from selective label correction, thus providing a dual correction guarantee mechanism for learning from numerical label noise. We use the estimated aleatoric uncertainties $\tilde{\sigma}_i^2$ for both refurbishable samples and identified clean samples as the variance parameters in the heteroscedastic loss. Under the heteroscedastic loss, DCGM assigns smaller sample weights to samples with false corrections and unidentified noisy samples. Obviously, these avoid noise accumulation caused by inaccurate label correction and noise identification effectively, and is beneficial for subsequent parameter updating, label noise identification and noisy label correction. Moreover, co-trained BNNs under the uncertainty consistency regularizer are adopted for ensuring the accuracy of aleatoric uncertainty estimation.

After adding the consistency regularizer, the total objective function of co-trained Bayesian neural networks is as follows:

$$L_{BNN} = L_{het} + L_{con}. \tag{5}$$

More specifically, the detailed formulas for the heteroscedastic regression loss and the consistency regularizer are as follows:

$$L_{het} = \frac{1}{n}\sum_{m=a,b}\sum_{i=1}^{n}\left(\frac{(y_i - \tilde{y}_{i,m})^2}{2\tilde{\sigma}^2_{i,m}} + \frac{\ln\tilde{\sigma}^2_{i,m}}{2}\right), \tag{6}$$

$$L_{con} = \frac{1}{n}\sum_{i=1}^{n}(\tilde{\sigma}^2_{i,a} - \tilde{\sigma}^2_{i,b})^2, \tag{7}$$

where $\tilde{\sigma}$ denotes predicted aleatoric uncertainty, the variables a and b represent two different Bayesian neural networks, respectively. Note that we predict the log-uncertainty $ln\sigma^2$ of each sample in practice, instead of aleatoric uncertainty σ^2, so as to avoid obtaining negative predictions for the variance.

DCGM utilizes the memory effect to guarantee the accuracy of initial noise identification and initial label correction [1]. Afterwards, by the data processing steps (neighborhood collective noise identification and label correction under the dual correction guarantee mechanism) and neural parameter updating, alternately, it updates neural parameters using sufficient and reliable data and finishes noise identification and label correction under well-tuned neural parameters in each epoch. As a result, its iterative process always moves towards better network parameters, thereby effectively mitigating the influence of numerical label noise.

D. Algorithm Description

Algorithm 1 summarizes the whole procedure of proposed DCGM in detail. During the warm-up period, neural networks are pre-trained on the entire training data set. According to the memorization effect, only clean and simple patterns are learned in the warm-up period, which will facilitate subsequent noise verification and label correction [1]. Then, with the improved BNNs, DCGM identifies noisy samples and selectively corrects noisy labels based on (3) and (4). Afterwards, under the heteroscedastic loss, two Bayesian neural networks are co-trained using the clean set D_C and the refurbishable set D_{ref}, as shown in (1) and (5), this sample weighting process is utilized as the post-correction in DCGM. With the increase in training epochs, the generalization performance of co-trained BNNs and the accuracy of discriminative feature representations and data processing will improve progressively. Note that the refurbished samples are aggregated across epochs except for the warm-up period for reusing.

The major computation cost of DCGM is incurred on noise identification, label correction and network parameter updating. Let T_{BNN} denote the computational cost of one forward-backward iteration for a single sample using the BNN. Let p^* be the dimension of z_i, and let $\tilde{m}$ denote the number of discrete $\tilde{y}_i$ points sampled for GMM fitting and MAP-based true label estimation. In experiments, we set $\tilde{m} = 1000$. Then, the cost of noise identification is $O(nT_{BNN} + n^2p^* + nK + nK\tilde{m})$. For selective label

Algorithm 1: DCGM Algorithm.

```
Input: Noisy data: D, epochs: Q, the batch size: m,
  warm-up epochs: τ*, uncertainty threshold: ε
Output: Optimal BNNs: θ_{t,m}
t = 1;
▲ → ∅; /* ▲ represents entire noisy samples*/
θ_{t,a} and θ_{t,b} ← Initialize the network parameters;
for i* = 1 to Q do
  if i* < τ* then
    WarmUp (D; θ_a, θ_b); /* Warm-up periods*/
  else
    Identify noisy samples according to (3) on D;
    Update ▲; /*Aggregation*/
    for each x_i ∈ D_N or x ∈ ▲ do
      if PU(x_i; q) ≤ ε then
        Selectively correct label noise by (4);
      end if
    end for
    Determine D_C and D_ref;
    Update network parameters θ_{t+1,a} and θ_{t+1,b};
  end if
  t = t + 1;
end for
return Bayesian network parameters: θ_{t,a} and θ_{t,b}.
```

correction, the computational complexity is $O(n_{\blacktriangle}T_{BNN} + n_{\blacktriangle}q + n_{D_{ref}}q)$. The time cost of pose-correction and network parameter updating is $O(n_{D_C \bigcup D_{ref}}T_{BNN})$. Hence, the total computational cost of Algorithm 1 is $O(\zeta(n^2p^* + nK\tilde{m} + n_{\blacktriangle}q + max(n, n_{D_C \bigcup D_{ref}})T_{BNN}))$ if there are ζ epochs.

E. Theoretical Analysis

Lemma 1: [29] In regression tasks under the squared loss, for any $\delta \in (0,1)$, the following generalization error bound holds with probability at least $1-\delta$:

$$R(f,\hat{D}) \leq \hat{R}(f,\hat{D}) \cdot \eta(h,n,\delta), \tag{8}$$

where $R(f,\hat{D})$ represents the expected risk of learner f trained on any regression data set $\hat{D}$, $\hat{R}(f,\hat{D})$ denotes the empirical risk, and $\eta(h,n,\delta) = (1-\sqrt{\frac{h(\ln(n/h)+1)-\ln\delta)}{n}})_+^{-1}$ depends on the VC dimension h.

Theorem 1: For regression learning on any noisy data set D, increasing the sample size while reducing the noise level can tighten the generalization error bound of the learner f with the squared loss.

Proof: Let $r_i = f(x_i) - y_i$. In noisy regression tasks, the true expression of empirical risk is:

$$\begin{aligned}\hat{R}(f,D) &= \frac{1}{n}\sum_{x_i \in D}[f(x_i) - y_i^*]^2 \\ &= \frac{1}{n}\sum_{x_i \in D}[f(x_i) - y_i + \varepsilon_i]^2 \\ &= \frac{1}{n}\sum_{x_i \in D}[(f(x_i) - y_i)^2 + \varepsilon_i^2 + 2\varepsilon_i \cdot r_i]. \end{aligned} \tag{9}$$

Considering that the objective of the learner f is to minimize the squared error loss, under sufficient training, the first-order optimality condition at convergence gives $E_D(f(x_i) - y_i) = E_D(r_i) = 0$ (the model zero-residual assumption). In other words, Mean Squared Error-optimal predictors are unbiased with respect to the noisy training labels. When D is sufficiently large, the law of large numbers ensures $E_D(f(x_i) - y_i) = E_D(r_i) \approx 0$. Then,

$$\begin{aligned}\hat{R}(f,D) &= E_D[(f(x_i) - y_i)^2] + E_D(\varepsilon_i^2) + 2E_D(\varepsilon_i \cdot r_i) \\ &= E_D[(f(x_i) - y_i)^2] + E_D(\varepsilon_i^2) + 2Cov(\varepsilon_i, r_i),\end{aligned} \tag{10}$$

where $Cov(\varepsilon_i, r_i) = \rho(\varepsilon_i, r_i)\sqrt{E_D(\varepsilon_i^2)Var(r_i)}$.

After substituting (10) into Lemma 1, we have, the generalization error bound of the regression learner f with the squared loss in noisy regression tasks:

$$R(f,D) \leq \{E_D(r_i^2) + E_D(\varepsilon_i^2) + 2Cov(\varepsilon_i, r_i)\} \cdot \eta(h,n,\delta). \tag{11}$$

Because $E_D(r_i^2)$ and $Var(r_i)$ are fixed for a given noisy data set, the above generalization error bound of the learner f mainly depends on $E_D(\varepsilon_i^2)$ and $\eta(h,n,\delta)$. Obviously, reducing the noise-related term: $E_D(\varepsilon_i^2)$ and the sample size-related term $\eta(h,n,\delta)$ at the data level can tighten the above generalization error bound.

It is noted that under a given hypothesis set, the VC dimension h remains constant across various regression data sets owing to label independence. Hence, the function $\eta(h,n,\delta) = (1-\sqrt{\frac{h(\ln(n/h)+1)-\ln\delta)}{n}})_+^{-1}$ solely depends on the sample size. The larger sample size necessarily results in smaller $\eta(h,n,\delta)$.

Accordingly, the generalization error bound for noisy regression tasks depends solely on $E_D(\varepsilon_i^2)$ and n. In summary, increasing the sample size while reducing the noise level can tighten the generalization error bound of the learner f with the squared loss on noisy regression data sets, which is the ultimate goal for learning from numerical label noise.

This means that selective label correction with high precision, after implementing accurate noise identification at the data level, can effectively reduce the generalization error bound for noisy regression tasks. ■

By Theorem 1, if DCGM can lead to a lower noise level and a larger sample size, it will yield a lower generalization error bound for learning from numerical label noise, thereby enabling a robust learning for the training data.

Theorem 2: Let Υ_p denote the confirmation bias from proposed neighborhood collective noise identification, and Υ_i represent the confirmation bias in noise identification based on predicted label distributions of individual samples. For any sample of the noisy data set D, $\Upsilon_p < \Upsilon_i$.

Proof: Let $|\Delta_i|$ denote the difference between the true output y_i^* and the model prediction $\tilde{y}_i$, i.e., $\tilde{y}_i = y_i^* + \Delta_i$. In learning from noisy labels, noise verification using the predicted label distributions of individual samples detects noisy samples via the absolute difference $|\tilde{y}_i - y_i|$. Differently, the proposed neighborhood collective noise identification recognizes noisy samples according to $|y_i - V(x_i, y_i)|$, where $V(x_i, y_i)$ is the neighborhood-based label consistency baseline.

For the sake of convenience in proof, let $V(x_i, y_i) = \sum_{k=1}^{K} \pi_k \tilde{y}_i^{(k)}$, where $\tilde{y}_i^{(k)}$ represents the model prediction for the $k-$th genuine nearest neighbor of the sample pair (x_i, y_i), then

$$\begin{aligned}&|y_i - [\pi_1 \tilde{y}_i^{(1)} + \cdots + \pi_K \tilde{y}_i^{(K)}]| \\ &= |y_i - \left[\pi_1 {y_i}^{*(1)} + \pi_1 \Delta_i^{(1)} + \cdots + \pi_K {y_i}^{*(K)} + \pi_K \Delta_i^{(K)}\right]| \\ &= |y_i - [\pi_1 {y_i}^{*(1)} + \cdots + \pi_K {y_i}^{*(K)}] - \varpi_i|, \end{aligned} \tag{12}$$

where $\varpi_i = \pi_1 \Delta_i^{(1)} + \cdots + \pi_K \Delta_i^{(K)}$.

Note that genuine neighbor finding has high accuracy on nearest neighbor search. Considering that the genuine nearest neighbors in the mapping feature space have very similar true numerical outputs with the sample pair (x_i, y_i), we have $|\pi_1 {y_i}^{*(1)} + \cdots + \pi_1 {y_i}^{*(K)} - y_i^*| \to 0$.

Then, compared with $|y_i - \tilde{y}_i| = |y_i - y_i^* - \Delta_i|$, which may introduce a larger deviation $|\Delta_i|$ for precise noise verification, the proposed neighborhood collective noise identification is more reliable and stable, because of neighborhood collective noise identification and the weighted mean strategy on the deviation $|\Delta_i|$ of each nearest neighbor. Evidently, the proposed noise identification can mitigate the impact of certain inaccurate model predictions to some extent, thereby helping to avoid confirmation bias of noise identification. In summary, $\Upsilon_p < \Upsilon_i$. ■

Theorem 2 implies that our neighborhood collective noise identification can alleviate the confirmation bias of common noise identification methods for numerical label noise to some extent. Hence, DCGM can relatively accurately distinguish noisy samples from clean ones.

Theorem 3: For any noisy data set D, BNNs co-trained under the heteroscedastic loss yield lower generalization error bounds in comparison with the general regression learner f with the squared loss:

$$\begin{aligned} &R(f, D) \leq \hat{R}(f, D) \cdot \eta(h, n, \delta), \\ &R(f_{BNN}, D) \leq \hat{R}(f_{BNN}, D) \cdot \eta(h, n, \delta), \\ &\hat{R}(f_{BNN}, D) \cdot \eta(h, n, \delta) < \hat{R}(f, D) \cdot \eta(h, n, \delta), \end{aligned} \tag{13}$$

where f_{BNN} denotes the BNN.

Proof: By proof of Theorem 1, for noisy regression tasks, we have

$$R(f, D) \leq \{E_D(r_i^2) + E_D(\varepsilon_i^2) + 2E_D(\varepsilon_i \cdot r_i)\} \cdot \eta(h, n, \delta).$$

Suppose that numerical label noises follow different noise distributions to better simulate the complex realistic noise environment, i.e., $p(\varepsilon_i) \sim N(\varepsilon_i | 0, \sigma_i^2)$. According to the M-estimation theory, there is a corresponding relationship between the form of the loss function and the noise distribution [12]. By maximum likelihood estimation, the heteroscedastic regression loss is more robust for the above realistic noise assumption:

$$\max_\theta \log p(y|x, \theta) \to L_{het} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{(y_i - \tilde{y}_i)^2}{2\tilde{\sigma}_i^2} + \frac{\ln \tilde{\sigma}_i^2}{2} \right). \tag{14}$$

Within the co-training framework, BNNs can derive a reasonably accurate estimate of the noise variances, which ensures the practicality and effectiveness of the heteroscedastic regression loss [11]. During network parameters learning, BNNs co-trained under the robust heteroscedastic regression loss will precisely assign smaller sample weights to noisy labels, thereby effectively resisting the influence of label noise.

In this case, the expression of the true empirical risk of noisy regression tasks will become:

$$\begin{aligned} \hat{R}(f_{BNN}, D) &= \frac{1}{n} \sum_{x_i \in D} [f_{BNN}(x_i) - y_i^*]^2 \\ &= \frac{1}{n} \sum_{x_i \in D} w_i [f(x_i) - y_i^*]^2 \\ &= \frac{1}{n} \sum_{x_i \in D} w_i [f(x_i) - y_i + \varepsilon_i]^2 \\ &= E_D[w_i r_i^2] + E_D(w_i \varepsilon_i^2) + 2E_D(w_i \varepsilon_i \cdot r_i) \end{aligned} \tag{15}$$

Given sample weights $w_i \in [0, 1]$, $E_D[w_i(f(x_i) - y_i)^2] + E_D(w_i \varepsilon_i^2) + 2E_D(w_i \varepsilon_i \cdot r_i) < E_D[(f(x_i) - y_i)^2] + E_D(\varepsilon_i^2) + 2E_D(\varepsilon_i \cdot r_i)$ on noisy regression data sets. Then, $\hat{R}(f_{BNN}, D) < \hat{R}(f, D)$.

In conclusion, $\hat{R}(f_{BNN}, D) \cdot \eta(h, n, \delta) < \hat{R}(f, D) \cdot \eta(h, n, \delta)$, i.e., BNNs co-trained under the heteroscedastic loss yield lower generalization error bounds in comparison with the general regression learner f with the squared loss. ■

From Theorem 3, the heteroscedastic loss can implicitly refurbish corrected labels by assigning smaller weights to samples with false corrections and unidentified noisy samples, thereby avoiding the impact of inaccurate label correction on model performance. Inspired by this, DCGM utilizes the heteroscedastic loss as a post-correction guarantee measure following selective label correction, and presents a dual correction guarantee mechanism for learning from noisy labels. Obviously, DCGM can achieve effective data expansion for regression tasks through numerical label correction under the dual correction guarantee mechanism, thereby enabling more robust updates for network parameters.

After selectively correcting identified label noises with high precision from the noisy set (which is aggregated across epochs except for warm-up periods), the training data set of DCGM will have a larger sample size. Fig. 2 presents a practical case illustrating the proportions of clean labels and label noise before and after label correction by DCGM. According to Theorems 2 and 3, the noise level of the training data set of DCGM is lower. Hence, the generalization ability of proposed DCGM is guaranteed, due to the dual correction guarantee mechanism.

IV. Experiments

A. Data Sets and Experimental Setup

1) Real-World Datasets: We verify the validity of the proposed method on several real-world regression data sets, including nine UCI data sets [30], the image data set for age

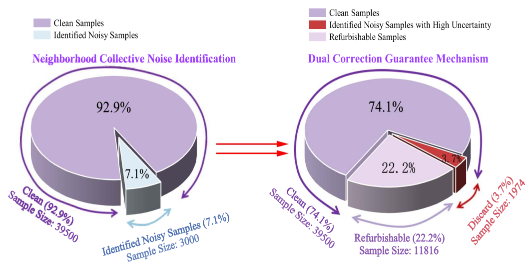


Fig. 2. Comparison of clean label and label noise proportions before and after label correction by DCGM on IMDB-WIKI (an image data set for age estimation that contains approximately 8% real label noise). Under the dual correction guarantee mechanism, IMDB-WIKI will own a larger sample size 51316, with very few noisy labels remaining.

TABLE I
UCI REGRESSION DATA SETS

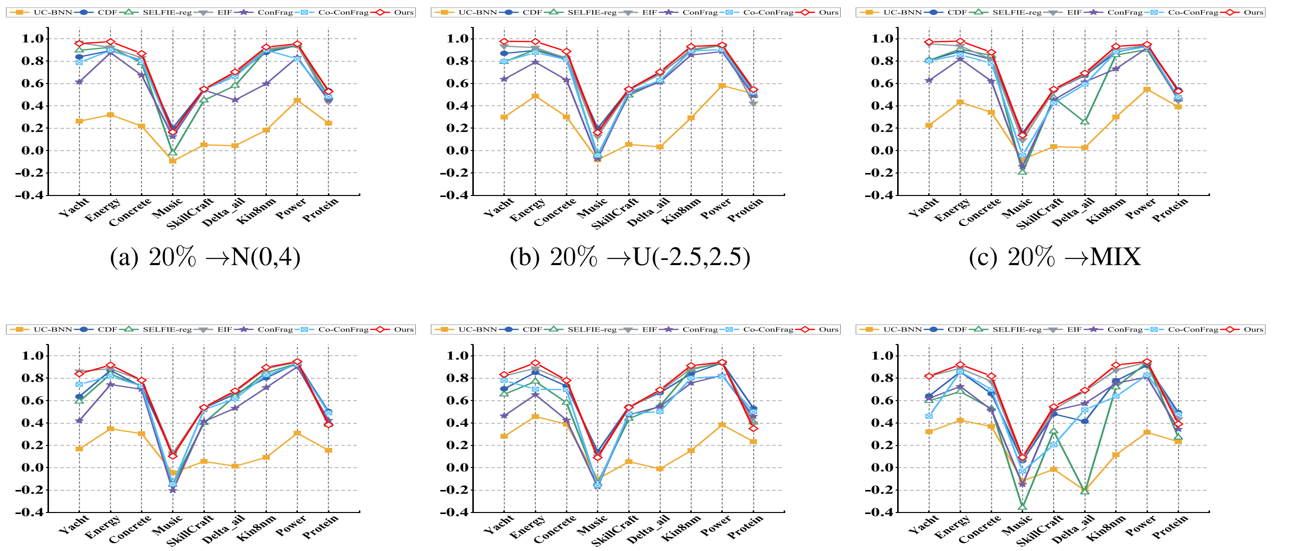
Real Data Sets	♯Samples	♯Features
Yacht	308	6
Energy	768	8
Concrete	1030	9
Music	1059	68
SkillCraft	3338	18
Delta_ail	7129	5
Kin8nm	8192	8
Power	9568	4
Protein	45730	9

estimation: IMDB-WIKI [31], and the facial image data set for gaze estimation: MPIIGaze [32]. Among them, IMDB-WIKI and MPIIGaze contain real-world label noise due to individual differences, environmental conditions, and technical limitations in the automatic web crawling process. Based on our observation, the noise in IMDB-WIKI exhibits a certain degree of skewness and asymmetry. Given that it is unknown whether UCI data sets contain label noise or not, three types of label noise are randomly added to training sets, respectively, i.e., Gaussian noise $N(0,4)$, uniformly distributed noise over $[-2.5, 2.5]$, mixture noise $50\%N(1,1) + 50\%U(-2.5, 2.5)$, with noise ratios 20% and 40%. Note that we did not apply any data augmentation technique on above data sets.

- *UCI data sets:* First, 9 UCI regression data sets are used to demonstrate the effectiveness of the proposed method. Table I lists these 9 UCI data sets. Note that all UCI data are column-wise normalized into the interval $[-1, 1]$. We adopt mean absolute error (MAE) (↓) and R^2 (↑) for evaluating the method performance. In addition, to evaluate experimental results convincingly, we apply 5-fold cross-validation on each UCI regression data set.
- *Age estimation. IMDB-WIKI:* In age estimation from photographs, we use a total of 50000 facial images from the IMDB-WIKI dataset, as provided in Ref. [31], to conduct experiments. The train-test splitting strategy follows that of the prior work (Ref. [31]), with 42500 images used for training and the remaining 7500 images for testing. The age labels of IMDB-WIKI range from 15 to 65. Note that through human inspection, IMDB-WIKI contains real-world label noise with an approximate rate of 8%. Moreover, we did not perform any additional image preprocessing on IMDB-WIKI. The evaluation index is utilized is the mean absolute error (MAE) (↓).
- *Gaze estimation. MPIIGaze:* For gaze estimation from facial photographs, we evaluate the effectiveness of DCGM on the most widely used data set, MPIIGaze, which comprises 213659 face images from 15 individuals captured over several months in their daily lives. These face images cover a wide range of different lighting conditions, eye appearances, and head postures [32]. The gaze direction of each person g $\in R^3$, which is composed of the horizontal yaw angle and the vertical pitch angle, is the label for each image. We preprocess these face images according to the state-of-the-art gaze estimation method L2CS-Net [33]. For the train-test split, following the experimental setup of L2CS-Net, we use 15-fold leave-one-subject-out cross-validation to evaluate the generalization ability of DCGM effectively. Likewise, the gaze angular error (i.e., $\frac{\mathrm{g}\cdot\hat{\mathrm{g}}}{\|\mathrm{g}\|\|\hat{\mathrm{g}}\|}$, ↓) is adopted as the evaluation index, where $\hat{\mathrm{g}} \in R^3$ denotes the predicted gaze direction vector.

2) Settings: For UCI data sets, MLP with three hidden layers is adopted as the network structure of BNNs. The networks are trained for 600 epoches (15 epoches in warm-up). Following the experimental setup of Ref. [11], we select ResNet-50 as the backbone network for image data sets, training it for 40 epochs (including 10 warm-up epochs) in age estimation and 20 epochs (including 5 warm-up epochs) in gaze estimation. We set the dropout probability to 2% for UCI datasets and 5% for image datasets, respectively, and use 5 Monte Carlo samples (i.e., $\tilde{T} = 5$) for variational inference. Regarding the hyperparameters, we set the number of nearest neighbors $K = 5$, perform cross-validation to determine the threshold τ, and conduct a grid search within $\{0.005, 0.01, 0.015, 0.02, 0.05, 0.1\}$ to find the best predictive uncertainty threshold. The grid search range for τ is determined based on the noise rate. Specifically, we first define the range of the noise rate for grid search. Then, the deviations $d_i = |y_i - V(x_i, y_i)|$ of training samples in first epoch for noise identification are sorted in descending order. Finally, the minimum values of the deviations that fall within the top portion corresponding to each predefined noise ratio are adopted to construct the grid search range for the threshold τ. After evaluation, we finally determine the optimal uncertainty thresholds: 0.01 for UCI and IMDB-WIKI, and 0.1 for MPIIGaze. More details regarding the model parameter settings are provided as follows.

- *UCI data sets:* We use the AdamW optimizer for training with a learning rate of 0.001, a batch size of 256 or 64 (depending on the sample size), a weight decay of 0.0001.
- *Age estimation. IMDB-WIKI:* We use the AdamW optimizer for training with a learning rate of 0.001, a weight decay of 0.0001, a batch size of 64. Note that the output of 1x1 convolutional layer ($512 \times$ block.expansion $\rightarrow 128$) is adopted to represent discriminative features. In addition,



(a) 20% →N(0,4) (b) 20% →U(-2.5,2.5) (c) 20% →MIX

(d) 40% →N(0,4) (e) 40% →U(-2.5,2.5) (f) 40% →MIX

Fig. 3. Comparison of R^2 (↑) on UCI data sets. Experimental results under the 20% noise rate are shown in Fig. 3(a)–(c). The consistent experimental results can be acquired at the 40% noise ratio, as shown in Fig. 3(d)–(f).

each experiment is conducted three times, and the mean results are reported with standard deviations.

- *Gaze estimation: MPIIGaze:* We use the AdamW optimizer for training with a learning rate of 0.00001, a weight decay of 0.0001, and a batch size of 16. The method for obtaining discriminative features is consistent with that for age estimation.

3) Baselines: The compared label noise learning methods include traditional machine learning approaches (CDF [4] and EIF [20]) and deep learning approaches (UC-BNN [11], SELFIE-reg [23], ConFrag [31], Co-ConFrag [31]). UC-BNN co-trains BNNs under the robust heteroscedastic regression loss. SELFIE-reg is a regression variant of SELFIE (a label noise learning method for classification). ConFrag and Co-ConFrag transforms regression tasks into classification problems before learning from noisy labels. Among the comparative methods mentioned above, Ref. [31] shows that Co-ConFrag outperforms fifteen advanced baselines in age estimation with noisy labels. Therefore, we adopt Co-ConFrag as the strongest baseline on IMDB-WIKI. Other baselines for IMDB-WIKI are CNLCU-S,H [7], Sigua [34], SPR [35], BMM [36], DY-S [36], SuperLoss [37], C-mixup [23], RDI [38], CDR [39], D2L [40], AUX [38], SELFIE [15], and Co-SELFIE [15].

As the state-of-the-art approach for gaze estimation, the L2CS-Net method [33] is adopted as the baseline on MPIIGaze. DCGM is integrated into the L2CS-Net framework to provide noise-addressing operations, thereby enabling the robust accomplishment of the gaze estimation task. Compared with the standard DCGM, the main differences lies in their objective functions and data preprocessing operations. The objective function of DCGM for gaze estimation additionally incorporates the cross-entropy loss.

All experiments were conducted on NVIDIA A800 GPUs.

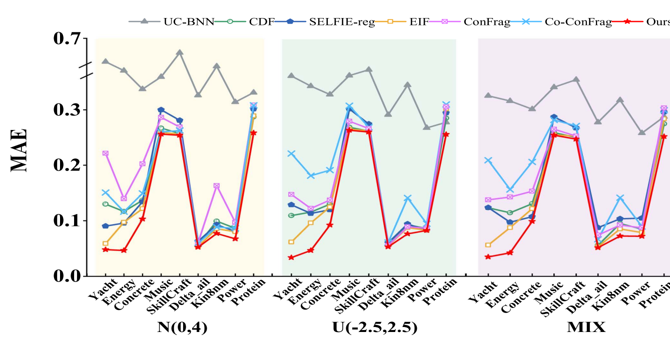


Fig. 4. Comparison of MAE (↓) on UCI data sets under the 20% noise rate. On data sets **Music**, **SkillCraft** and **Delta_ail**, DCGM is slightly more robust than EIF, which ranks second on the MAE metric (See Table II for detailed experimental results). Both of the above methods attain the best performance on **Power** with uniform noise $U(-2.5, 2.5)$.

B. Experimental Results on UCI Data Sets

1) Experimental Results Under the 20% Noise Ratio: As shown in Figs. 3 and 4, across various noise types with the 20% noise rate, the proposed method consistently outperforms others on both MAE and R^2 metrics, even under the mixed noise, which is a more realistic noise assumption. Specifically, DCGM obtains the lowest MAE across all nine UCI data sets. Due to the visual ambiguity of these results presented in Fig. 4, we present the specific MAE values for the ambiguous cases in Table II to better demonstrate the superiority of our approach. Moreover, DCGM achieves the highest value of R^2 with the greatest frequency. We tallied how often each algorithm achieved the highest R^2 across nine data sets with three noise types. Our method ranks first with 21 occurrences, followed by CDF (with 5 occurrences) and EIF (with 1 occurrence). Note that above evaluation indexes are in decreasing order of priority when evaluating robustness. All the

TABLE II
SPECIFIC MAE VALUES AT VISUALLY AMBIGUOUS POSITIONS IN THE MAE COMPARISON UNDER DIFFERENT NOISE RATES (UCI)

Data sets	Methods	20% Noise Rate			40% Noise Rate		
		N(0,4)	U(-2.5,2.5)	MIX	N(0,4)	U(-2.5,2.5)	MIX
Music	EIF	0.2589	0.2603	0.2622	0.2654	0.2692	0.2677
	CDF	0.2669	0.2637	0.2662	0.2969	0.2856	0.2872
	Ours	**0.2558**	**0.2583**	**0.2613**	**0.2624**	**0.2646**	**0.2650**
SkillCraft	EIF	0.2561	0.2577	0.2600	0.2587	0.2591	0.2616
	CDF	0.2570	0.2569	0.2576	0.2613	0.2587	0.2719
	Ours	**0.2542**	**0.2552**	**0.2542**	**0.2574**	**0.2567**	**0.2549**
Delta_ail	EIF	0.0540	0.0540	0.0542	0.0544	0.0545	0.0543
	CDF	0.0569	0.0560	0.0571	0.0590	0.0580	0.0848
	Ours	**0.0523**	**0.0523**	**0.0528**	**0.0534**	**0.0529**	**0.0531**

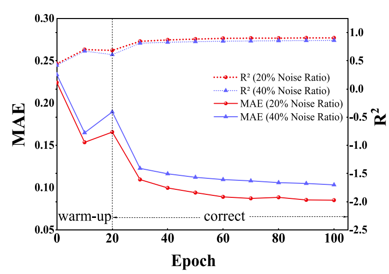


Fig. 5. Evolution of MAE (↓) and R^2 (↑) for testing data across training epochs on the noisy Delta_ail data set (UCI).

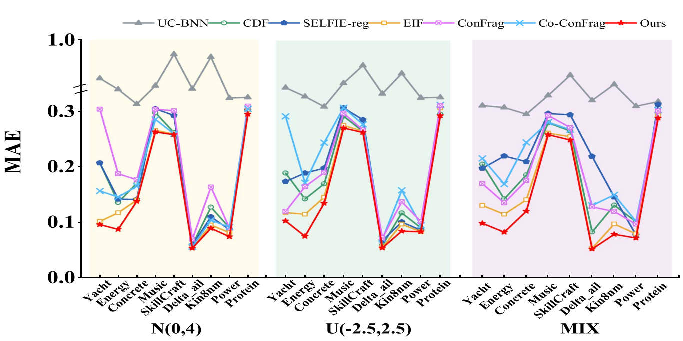


Fig. 6. Comparison of MAE on UCI data sets with the 40% noise rate.

above experimental results verify that DCGM can effectively resist label noise in the numerical form.

In addition, Fig. 5 depicts the evolution of MAE and R^2 for testing data across training epochs. As the iterations progress, DCGM exhibits an increasing capability to identify label noise and correct label noise, leading to a notable enhancement on model robustness.

2) Experimental Results Under the 40% Noise Ratio: As illustrated in Fig. 6, across different noise types at a 40% noise rate, the proposed method consistently outperforms other approaches on MAE, similarly to experimental results under the 20% noise ratio, even in mixed noise scenarios. This reveals that the proposed DCGM can effectively mitigate the impact of numerical label noise, irrespective of the noise ratio. In data sets **Music**, **SkillCraft** and **Delta_ail**, DCGM is slightly better than EIF, which ranks second on the MAE metric. The specific MAE values for the comparison of these visually ambiguous experimental results are presented in Table II. In addition, both

TABLE III
COMPARISON OF PERFORMANCE FOR NOISY LABEL DETECTION AND LABEL CORRECTION ON DELTA_AIL WITH MIXTURE NOISE

Methods	Delta_ail_MIX_20%				Delta_ail_MIX_40%			
	Accuracy	Recall	Corr. MAE	Orig. MAE	Accuracy	Recall	Corr. MAE	Orig. MAE
EIF	80.44%	93.42%	-	-	90.99%	93.91%	-	-
CDF	77.65%	75.26%	-	-	90.54%	65.06%	-	-
Co-ConFrag	85.79%	60.30%	-	-	84.21%	58.59%	-	-
Ours	**90.00%**	**98.09%**	**0.0537**	**1.306**	**91.03%**	**99.09%**	**0.053**	**1.2964**

DCGM and EIF achieve the best performance on **Power** with uniform noise $U(-2.5, 2.5)$.

Fig. 3 shows the R^2 comparison of each method across nine UCI data sets under various noise conditions with a 40% noise rate. We tallied how often each algorithm achieved the highest R^2 across nine data sets with three noise types. Our method ranks first with 19 occurrences, followed by CDF (5 times) and EIF (3 times). Generally speaking, the proposed method is the most robust in terms of the R^2 metric at a 40% noise ratio. All these experimental results on UCI data sets clearly demonstrate that, regardless of the distribution and the ratio of label noise, the proposed DCGM has better robustness in comparison with comparative methods in learning from noisy numerical labels.

Besides, Table III presents a performance comparison of noisy label detection and label correction. Corr. MAE and Orig. MAE are abbreviations for $\frac{1}{\breve{n}}\sum_{i=1}^{\breve{n}}|y_i^* - y_i^{correct}|$ and $\frac{1}{\breve{n}}\sum_{i=1}^{\breve{n}}|y_i - y_i^*|$, respectively, where $\breve{n}$ denotes the number of noisy samples that are selectively corrected. Compared with three best comparative methods in previous experiments, DCGM achieves the highest accuracy and recall for noise recognition. Since no dedicated label correction methods exist for regression tasks, we only report the correction results of our method. From Table III, after our selective correction, the corrected labels are much closer to the ground-truth labels (Corr. MAE < Orig. MAE). Furthermore, the implicit correction mechanism of DCGM assigns smaller weights to falsely corrected samples during model training. Obviously, the proposed noise identification and label correction module can effectively reduce the noise level as the sample size increases.

C. Experimental Results in Age Estimation

Table IV depicts the MAE comparison on the real noisy image data set IMDB-WIKI for age estimation. From Table IV, DCGM acquires the best performance **6.24**. Additionally, we present some examples of noisy age labels identified and corrected by DCGM in Table V. Here, deviation refers to the difference between the original age label and the corrected age label. From Table V, DCGM effectively identifies label noise and assigns them age labels that better match image descriptions, thus enhancing the method robustness. The above experimental results demonstrate that DCGM has good label correction ability for numerical label noise, its robust dual correction guarantee mechanism effectively overcomes the challenges in label correction for regression tasks.

DCGM occasionally misclassifies clean samples as noisy samples, even so, the method performance is still guaranteed, because it can correct age labels well as training epochs progress

TABLE IV
COMPARISON OF MAE ON THE REAL NOISY DATA SET IMDB-WIKI. THE EXPERIMENTAL RESULTS OF COMPARATIVE METHODS COME FROM REF. [31], WHICH PROPOSES CONFRAG AND CO-CONFRAG. AND THE NUMBER IN PARENTHESES DENOTES THE STANDARD DEVIATION.

Method	CNLCU-S	BMM	SELFIE	Sigua	RDI	Co-SELFIE	CDR	D2L	SPR	CNLCU-H	ConFrag	DY-S	Superloss	AUX	C-Mixup	Co-ConFrag	Ours
MAE	8.78 (0.16)	8.52 (0.13)	8.24 (0.92)	7.37 (0.12)	7.30 (0.04)	7.20 (0.46)	7.20 (0.01)	7.18 (0.12)	7.16 (0.03)	7.01 (0.02)	7.00 (0.13)	6.98 (0.07)	6.97 (0.05)	6.96 (0.05)	6.84 (0.04)	6.58 (0.06)	**6.24 (0.02)**
Rank	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	**1**

TABLE V
NOISY AGE LABELS IDENTIFIED AND CORRECTED BY DCGM ON IMDB-WIKI

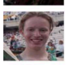
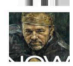
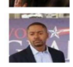
Image	Age Label	Label Correction	Deviation	Source	Image	Age Label	Label Correction	Deviation	Source
	15 ↓	19.20	4.20	imdb		25 ↓	30.17	5.17	imdb
	15 ↓	23.33	8.33	imdb		25 ↓	29.32	4.32	imdb
	15 ↓	19.81	4.81	wiki		34 ↓	40.29	6.29	imdb
	49 ↓	57.03	8.03	wiki		39 ↓	50.80	11.80	imdb
	16 ↓	25.51	9.51	imdb		63 ↑	53.28	9.72	imdb
	27 ↓	33.29	6.29	imdb		29 ↓	38.16	9.16	imdb

TABLE VI
LABEL CORRECTION BY DCGM FOR SAMPLES MISIDENTIFIED AS NOISY SAMPLES

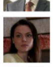
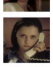
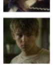
Image	Age Label	Label Correction	Deviation	Source
	53	52.78	0.22	imdb
	52	52.17	0.17	imdb
	25	25.26	0.26	imdb
	18	18.49	0.49	imdb
	21	21.12	0.12	imdb

TABLE VII
COMPARISON OF TRAINING TIME AND COMPUTATIONAL COST ON IMDB-WIKI

Methods	Total	Total Noise Identification	Parameters (M)	Size (MB)	Peak Memory (GB)
Co-ConFrag	4h 11min	2h 35min	23.56	89.99	25.41
Ours	4h 14min 50s	1h 18min 30s	24.56	93.69	18
Runtime of each module (per epoch)	Noise Identification	Label Correction	Post-correction + Forward-Backward		
	2min 37s	2min 41s	2min 30s		

TABLE VIII
PERFORMANCE EVALUATION ON SIMULATED REAL-WORLD LABEL NOISE

Methods	Kin8nm_skewed_20%		Kin8nm_skewed_40%		IMDB-Clean_IDN_20%	IMDB-Clean_structured_20%
	MAE	R^2	MAE	R^2	MAE	MAE
EIF	0.0875	0.9038	0.0972	0.8800	–	–
CDF	0.0898	0.9000	0.1069	0.8604	–	–
Co-ConFrag	0.0847	0.9132	0.1225	0.8164	8.09	7.63
Ours	**0.0759**	**0.9256**	**0.0874**	**0.9054**	**6.74**	**6.82**

Note: The location, skewness, scale, and degrees of freedom (DOF) parameters of the skewed noise from the GHSkewt distribution are 0, 0.8, 1, and 10, respectively.

TABLE IX
COMPARISON OF MEAN GAZE ANGULAR ERROR ON MPIIGAZE

Methods	MPIIGaze
FAR-Net [44]	4.3°
CA-Net [45]	4.1°
AGE-Net [46]	4.09°
L2CS-Net [33]	3.92°
Ours	**3.56°**

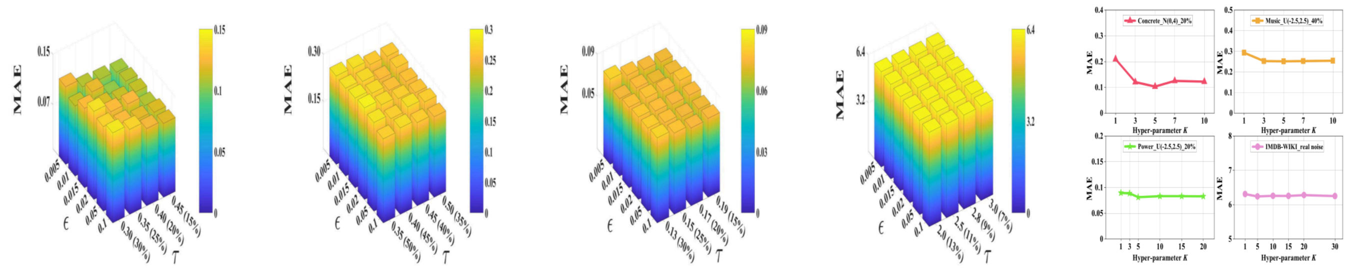
(see Table VI). This further demonstrates the effectiveness of our proposed dual correction guarantee mechanism. Under the robust neighborhood collective noise identification and dual correction guarantee mechanism, noise identification and label correction of the proposed DCGM is highly accurate. Therefore, it remains the most robust method, even on real-world noisy datasets.

Table VII presents a comparison of training time and computational cost on the large-scale IMDB-WIKI image dataset. Compared with the best-performing baseline Co-ConFrag (which does not perform label correction), DCGM incurs comparable runtime and memory overhead with shorter noise identification time, while achieving significantly lower MAE and stronger robustness against numerical label noise.

Finally, we also compare the performance of DCGM with the best-performing baselines under more complex noise scenarios: skewed noise, structured noise, and instance-dependent noise (IDN). These synthetic noise types can effectively simulate realistic noisy label settings. The construction methods are detailed in Refs. [31], [41], [42], [43]. From Table VIII, our method consistently achieves the best performance.

D. Experimental Results in Gaze Estimation

Table IX presents a comparison between DCGM and the state-of-the-art gaze estimation approach L2CSNet on MPIIGaze. The experimental results of comparative methods come from Ref. [33], which proposes L2CSNet. As shown, the proposed method achieves the lowest mean gaze angle error of **3.56°**, significantly outperforming L2CSNet. Table X shows some typical noisy gaze directions identified and corrected by DCGM. It is evident that, for noisy gaze labels on MPIIGaze, both the



(a) Concrete_N(0,4)_20% (b) Music_U(-2.5,2.5)_40% (c) Power_U(-2.5,2.5)_20% (d) IMDB-WIKI_real noise (e) number of nearest neighbors K

Fig. 7. Sensitivity analysis of hyperparameters τ, ϵ and K. The grid search range of hyperparameter τ is actually determined based on the noise rate (see the percentages in parentheses adjacent to the x-axis).

TABLE X
NOISY GAZE DIRECTIONS IDENTIFIED AND CORRECTED BY DCGM. IN FOLLOWING FACE IMAGES, RED ARROWS DENOTE ORIGINAL (NOISY) GAZE LABELS, GREEN ARROWS REPRESENT CORRECTED GAZE DIRECTIONS, AND BLUE ARROWS ARE PREDICTED GAZE DIRECTIONS FROM OUR FINAL NETWORKS.

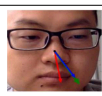
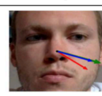
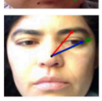
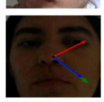
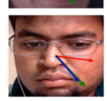
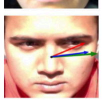
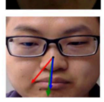
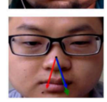
No.	Image	No.	Image	No.	Image
1		2		3	
4		5		6	
7		8		9	

TABLE XI
ABLATION STUDY ON IMDB-WIKI. THE "IDEN." AND "SELCORR." ARE ABBREVIATIONS FOR "IDENTIFICATION" AND "SELECTIVE LABEL CORRECTION", RESPECTIVELY. THE ROBUST BNNS REFER TO CO-TARINED BNNS UNDER THE HETEROSCEDASTIC OBJECTIVE LOSS.

Methods	Modules	MAE
robust BNNs	w/o Filtering + Correction	6.32
DCGM-f	w/o Correction	6.29
DCGM-all	all label noises are corrected	6.37
DCGM-o	w/o GMM + discriminative	6.33
DCGM-MLP	w/o post-correction	6.32
Ours	Noise iden. + SelCorr. + post-correction	**6.24**

corrected gaze directions and the predicted gaze directions from our final networks are closer to the ground-truth gaze directions than the original gaze labels. These visual comparisons convincingly demonstrate the effectiveness of DCGM in noisy regression tasks.

E. Sensitivity Analysis

The influence of hyperparameters on the model performance is investigated in this subsection. Without loss of generality, we conduct experiments on two types of data sets: the image data set IMDB-WIKI (which contains real-world noise) and several UCI data sets contaminated by various types of synthetic label noise under different noise ratios (see Fig. 7). We search the optimal hyperparameter ϵ in the range of $\{0.005, 0.01, 0.015, 0.02, 0.05, 0.1\}$. The grid search range for the noise rate (used to determine the candidate set for the hyperparameter τ) is indicated by the percentages in the parentheses adjacent to the x-axis of Fig. 7. Fig. 7(a)–(d) illustrates the MAE values across different hyperparameter combinations.

We observe that the influence of hyperparameter selection on model performance is less than 0.04 on UCI data sets and less than 0.1 on IMDB-WIKI. On each data set, the MAE gradually decreases as the noise rate corresponding to hyperparameter τ approaches the true noise rate of the data set. Meanwhile, the MAE reaches its minimum value when the parameter ϵ takes an appropriate value 0.01, and rises when ϵ is either too large or too small. Generally speaking, the model performance is affected by the hyperparameter selection, indicating the contribution of our neighborhood collective noise identification and dual correction guarantee mechanism to the model performance.

Besides, we further evaluate the model performance across different values of the number of nearest neighbors K on the above data sets. According to sample size, we analyze the parameter K from {1,3,5,7,10} on Concrete and Music, {1,3,5,10,15,20} on Power, and {1,5,10,15,20,30} on IMDB-WIKI. From Fig. 7(e), when the number of neighbors is 1, the performance of the method is the worst. The increase of the genuine nearest neighbor number enhances the model's performance. These results indicate the significance of our neighborhood collective noise identification. Moreover, the choice of K does not significantly influence the performance across all data sets (except for $K = 1$). When K is set to the most commonly used neighborhood size ($K = 5$), the DCGM model achieves nearly optimal performance.

F. Ablation Study

An ablation study is conducted in this section to verify the effect of different components of DCGM. We compare five weakened variants of DCGM on IMDB-WIKI, including: 1) BNNs are trained under the robust heteroscedastic loss (without filtering and correction), 2) DCGM-MLP, which uses an MLP with squared loss as the network body, 3) DCGM-f, which omits label correction, 4) DCGM-all, which corrects all label noise, 5) DCGM-o, which identifies label noise based on the mean of predicted values of ordinary nearest neighbors (nearest neighbors are found according to raw sample features). The experimental results are shown in Table XI. From Table XI, DCGM achieves the smallest MAE (**6.24**). The proposed neighborhood collective noise identification, selective label correction and the post-correction guarantee regime each contributes individually to noisy regression tasks. Their combination is more effective for mitigating the impact of numerical label noise.

V. Conclusion

In this paper, we propose a novel label noise correction method with a dual correction guarantee mechanism for noisy regression tasks, which can complement existing research on learning from noisy labels. DCGM owns theoretical guarantees. It can accurately identify label noise after fully exploiting the information from genuine nearest neighbors, and then the credible clean data sets can be acquired. Different from current methods with just a single noise addressing mechanism, DCGM selectively refurbishes noisy labels that can be corrected with high precision, thereby leading to a nearly full exploration of training data. Additionally, owing to the adoption of the post-correction guarantee regime, DCGM exhibits stronger robustness in label correction, network parameter updating, and even noise identification. It may provide a flexible data-model interactive-driven manner for learning with label noise. Future research will explore more effective and efficient numerical label correction frameworks, integrate DCGM with vision-language models, and extend it to noisy imbalanced regression tasks.

References

[1] H. Song, M. Kim, D. Park, Y. Shin, and J.-G. Lee, "Learning from noisy labels with deep neural networks: A survey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8135–8153, Nov. 2023.

[2] J. Shu, X. Yuan, D. Meng, and Z. Xu, "CMW-Net: Learning a class-aware sample weighting mapping for robust deep learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 11521–11539, Oct. 2023.

[3] J. Li, R. Socher, and S. C. Hoi, "DivideMix: Learning with noisy labels as semi-supervised learning," in *Proc. Int. Conf. Learn. Representations*, 2020, pp. 8923–8936.

[4] G. Jiang, W. Wang, Y. Qian, and J. Liang, "A unified sample selection framework for output noise filtering: An error-bound perspective," *J. Mach. Learn. Res.*, vol. 22, no. 1, pp. 1–66, 2021.

[5] G. Jiang, J. Zhang, X. Bai, W. Wang, and D. Meng, "Which is more effective in label noise cleaning, correction or filtering," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 12866–12 873.

[6] Y. Gu and H. Zou, "Sparse composite quantile regression in ultrahigh dimensions with tuning parameter calibration," *IEEE Trans. Inf. Theory*, vol. 66, no. 11, pp. 7132–7154, Nov. 2020.

[7] X. Xia et al., "Sample selection with uncertainty of losses for learning with noisy labels," in *Proc. Int. Conf. Learn. Representations*, 2022, pp. 23214–23236.

[8] G. Jiang and W. Wang, "An interpretable sample selection framework against numerical label noise," *Mach. Learn.*, vol. 114, no. 1, 2025, Art. no. 43.

[9] Z. Zhou, *Machine Learning*. Beijing, China: Tsinghua Univ. Press, 2016.

[10] J. Ren, M. Zhang, C. Yu, and Z. Liu, "Balanced MSE for imbalanced visual regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 7926–7935.

[11] W. Dai, X. Li, and K.-T. Cheng, "Semi-supervised deep regression with uncertainty consistency and variational model ensembling via Bayesian neural networks," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 7304–7313.

[12] Y. Guo, W. Wang, and X. Wang, "A robust linear regression feature selection method for data sets with unknown noise," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 31–44, Jan. 2023.

[13] Y. Guo and W. Wang, "A robust adaptive linear regression method for severe noise," *Knowl. Inf. Syst.*, vol. 65, no. 11, pp. 4613–4653, Nov. 2023.

[14] J. Li, G. Li, F. Liu, and Y. Yu, "Neighborhood collective estimation for noisy label identification and correction," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 128–145.

[15] H. Song, M. Kim, and J.-G. Lee, "SELFIE: Refurbishing unclean samples for robust deep learning," in *Proc. 36th Int. Conf. Mach. Learn.*, 2019, pp. 5907–5915.

[16] Y. Li, H. Han, S. Shan, and X. Chen, "DISC: Learning from noisy labels via dynamic instance-specific selection and correction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 24070–24079.

[17] D. Kim, K. Ryoo, H. Cho, and S. Kim, "SplitNet: Learnable clean-noisy label splitting for learning with noisy labels," *Int. J. Comput. Vis.*, vol. 133, no. 2, pp. 549–566, 2025.

[18] J. Shu et al., "Meta-Weight-Net: Learning an explicit mapping for sample weighting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 1919–1930.

[19] X. Rui, X. Cao, J. Shu, Q. Zhao, and D. Meng, "Learning to adapt unknown noise for hyperspectral image denoising," 2024, *arXiv:2301.06081v2*.

[20] C. Li, Z. Mao, and M. Jia, "A real-valued label noise cleaning method based on ensemble iterative filtering with noise score," *Int. J. Mach. Learn. Cybern.*, vol. 15, no. 9, pp. 4093–4118, 2024.

[21] J. Martıń, J. A. Sáez, and E. Corchado, "Tackling the problem of noisy IoT sensor data in smart agriculture: Regression noise filters for enhanced evapotranspiration prediction," *Expert Syst. Appl.*, vol. 237, 2024, Art. no. 121608.

[22] G. Jiang, Z. Li, and W. Wang, "Maximum a posteriori estimation and filtering algorithm for numerical label noise," *Appl. Intell.*, vol. 54, pp. 8841–8855, 2024.

[23] S.-H. Hwang, M. Kim, and S. E. Whang, "RC-Mixup: A data augmentation strategy against noisy data for regression tasks," in *Proc. 30th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2024, pp. 1155–1165.

[24] Z. Zhou and M. Li, "Semisupervised regression with cotraining-style algorithms," *IEEE Trans. Knowl. Data Eng.*, vol. 19, no. 11, pp. 1479–1493, Nov. 2007.

[25] J. Gawlikowski et al., "A survey of uncertainty in deep neural networks," *Artif. Intell. Rev.*, vol. 56, no. Suppl 1, pp. 1513–1589, 2023.

[26] Y. Wei, J. Ye, Z. Huang, J. Zhang, and H. Shan, "Online prototype learning for online continual learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 18764–18 774.

[27] J. Choi and K.-E. Kim, "MAP inference for Bayesian inverse reinforcement learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2011, pp. 1989–1997.

[28] A. W. Bowman and A. Adelchi, *Applied Smoothing Techniques for Data Analysis*. London, U.K.: Oxford Univ. Press Inc., 1997.

[29] V. Cherkassky, X. Shao, F. M. Mulier, and V. N. Vapnik, "Model complexity control for regression using VC generalization bounds," *IEEE Trans. Neural Netw.*, vol. 10, no. 5, pp. 1075–1089, Sep. 1999.

[30] K. Markelle, L. Rachel, and N. Kolby, "The UCI machine learning repository," 2017. [Online]. Available: http://archive.ics.uci.edu

[31] C. D. Kim, S. Moon, J. Moon, D. Woo, and G. Kim, "Sample selection via contrastive fragmentation for noisy label regression," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 127561–127609.

[32] X. Zhang, Y. Sugano, M. Fritz, and A. Bulling, "MPIIGaze: Real-world dataset and deep appearance-based gaze estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 1, pp. 162–175, Jan. 2019.

[33] A. A. Abdelrahman, T. Hempel, A. Khalifa, A. Al-Hamadi, and L. Dinges, "L2CS-Net: Fine-grained gaze estimation in unconstrained environments," in *Proc. 8th Int. Conf. Front. Signal Process.*, 2023, pp. 98–102.

[34] B. Han et al., "Forgetting may make learning with noisy labels more robust," in *Proc. 37th Int. Conf. Mach. Learn.*, 2020, pp. 4006–4016.

[35] Y. Wang, X. Sun, and Y. Fu, "Scalable penalized regression for noise detection in learning with noisy labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 346–355.

[36] E. Arazo, D. Ortego, P. Albert, N. O'Connor, and K. Mcguinness, "Unsupervised label noise modeling and loss correction," in *Proc. 36th Int. Conf. Mach. Learn.*, 2019, pp. 312–321.

[37] T. Castells, P. Weinzaepfel, and J. Revaud, "SuperLoss: A generic loss for robust curriculum learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 4308–4319.

[38] W. Hu, Z. Li, and D. Yu, "Simple and effective regularization methods for training on noisily labeled data with generalization guarantee," in *Proc. Int. Conf. Learn. Representations*, 2020, pp. 4023–4040.

[39] X. Xia et al., "Robust early-learning: Hindering the memorization of noisy labels," in *Proc. Int. Conf. Learn. Representations*, 2021, pp. 12802–12816.

[40] X. Ma et al., "Dimensionality-driven learning with noisy labels," in *Proc. 35th Int. Conf. Mach. Learn.*, 2018, pp. 3355–3364.
[41] X. Liu and X. Yang, "Identification of nonlinear state-space systems with skewed measurement noises," *IEEE Trans. Circuits Syst. I: Reg. Papers*, vol. 69, no. 11, pp. 4654–4662, Nov. 2022.
[42] P. Chen, J. Ye, G. Chen, J. Zhao, and P.-A. Heng, "Beyond class-conditional assumption: A primary attempt to combat instance-dependent label noise," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 11442–11450.
[43] K. Yi and J. Wu, "Probabilistic end-to-end noise correction for learning with noisy labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 7017–7025.
[44] Y. Cheng, X. Zhang, F. Lu, and Y. Sato, "Gaze estimation by exploring two-eye asymmetry," *IEEE Trans. Image Process.*, vol. 29, pp. 5259–5272, 2020.
[45] Y. Cheng, S. Huang, F. Wang, C. Qian, and F. Lu, "A coarse-to-fine adaptive network for appearance-based gaze estimation," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 10623–10630.
[46] P. Biswas et al., "Appearance-based gaze estimation using attention and difference mechanism," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 3143–3152.

Yaqing Guo received the PhD degree from the School of Computer and Information Technology, Shanxi University, Taiyuan, China, in 2023. Currently she is a lecturer with the School of Computer and Information Technology, Shanxi University. Her research focuses on machine learning.

Lingxuan Cui is currently working toward the master's degree with the School of Computer and Information Technology, Shanxi University, Taiyuan, China. His research interests include machine learning and label noise learning.

Gaoxia Jiang received the PhD degree from the School of Computer and Information Technology, Shanxi University, Taiyuan, China, in 2018. He is currently an associate professor with the School of Computer and Information Technology, Shanxi University. His research interests include machine learning and data mining.

Fan Lei is currently working toward the master's degree with the School of Computer and Information Technology, Shanxi University, Taiyuan, China. His research interests include machine learning and label noise learning.

Senyu Hou is currently working toward the PhD degree with the School of Computer and Information Technology, Shanxi University, Taiyuan, China. His research interests include machine learning, weakly supervised learning, and label noise learning.

Hang Xu received the PhD degree from the School of Computer and Information Technology, Shanxi University, Taiyuan, China, in 2018. She is currently an associate professor with the School of Computer and Information Technology, Shanxi University, and serves as a master's supervisor in computer science and technology. Her current research interests include data mining, machine learning, service computing, etc.

Wenjian Wang received the BS degree in computer science from Shanxi University, China, in 1990, the MS degree in computer science from Hebei Polytechnic University, China, in 1993, and the PhD degree in applied mathematics from Xian Jiaotong University, China, in 2004. She is currently a full-time professor and the PhD supervisor with Shanxi University. Since 1993, she has been with the School of Computer and Information Technology, Shanxi University, where she was an associate professor, in 2000 and a full-time professor, in 2004. She has authored or coauthored more than 200 academic papers. Her current research interests include machine learning, data mining and computational intelligence.